

Flex Bridge Quick Start

FlexRadio 6000/8000 → QMAP / WS-MAP / WSJT-X, over the network. Version 3.1.3 (beta), N6NU - part of the SDR Bridges V3 family.

In short: start SmartSDR as usual, then start Flex Bridge. It finds the radio, attaches to SmartSDR, and sends the panadapter's IQ to QMAP on port 50004, labelled with the *exact* panadapter centre. QMAP needs no Fadd and no manual frequency. There's nothing to configure.

1. Use it with SmartSDR (recommended)

1. Start **SmartSDR** and connect to your radio as usual. Put the panadapter where you want QMAP to look, for example centred on 432.100 for the 432 EME segment.
2. Start **Flex Bridge**. The log and Settings show "*attached to SmartSDR station ...*". If you have more than one Flex, it asks which one, once.
3. Start **QMAP**: input on UDP port (the default), IQ sample rate , .

QMAP gets **96 kHz** around SmartSDR's panadapter centre, the rate every QMAP understands, including the stock one that comes with WSJT-X. If you drag or retune the panadapter in SmartSDR, QMAP's frequencies follow automatically. With WS-MAP (or the QMAP from WSJT-X Improved / WS) you can widen it to 192 kHz under . Start or restart QMAP *after* the bridge so its sample rate picks up the change.

With a transverter (144 / 222 / 432 / 1296)

- Define the transverter in SmartSDR, as most VHF/UHF Flex stations already do: **Settings** → **Transverters**, RF frequency (e.g.), IF (e.g.), XVTR port. Then select that band. SmartSDR shows the panadapter and slices in **RF**, e.g. 432.100, and the radio converts to the IF itself.
- The bridge passes that RF frequency to QMAP, so QMAP shows 432.xxx with . The bridge log confirms it on its first line: "*pan ... @ 432.100000 MHz via transverter ...*".
- WSJT-X through the bridge (section 2) also works in RF: the dial reads 432.065, and the radio transmits on the transverter's IF through the XVTR port, exactly as SmartSDR does.
- If you tune the **IF** instead, without a transverter entry in SmartSDR, the panadapter reads 28.100 and so does QMAP. Then set QMAP to the transverter's LO, e.g. for 432 on a 28 MHz IF. Defining the transverter in SmartSDR is simpler.

Polarisation H+V on two transverters with two bridges and two QMAPs is coming. This beta streams the panadapter that holds SmartSDR's TX slice.

The bridge leaves SmartSDR's panadapter, antenna, gain, power and DAX settings alone. Keep using WSJT-X through SmartSDR's CAT and DAX exactly as before if you like.

2. Optional: WSJT-X / WS through the bridge (TCI)

This replaces SmartSDR CAT, DAX audio and virtual cables with one connection:

WSJT-X setting	Value
Settings → Radio → Rig	TCI Client RX1, server 127.0.0.1:40001
Settings → Radio → PTT method	CAT
Settings → Radio → Split	None (split is not mapped yet)
Settings → Audio	TCI audio (input and output)

Click **Test CAT**: it turns green. WSJT-X then tunes SmartSDR's TX slice, and tuning the slice in SmartSDR moves WSJT-X. RX audio is the radio's own DAX audio of that slice. TX audio and PTT go to the radio as DAX TX. The bridge sets the slices it drives to DIGU (USB data).

A second WSJT-X on server 127.0.0.1:40002 gets a real Flex slice, B: SmartSDR's own second slice if there is one, otherwise one the bridge adds and removes again on exit. It has its own DAX audio and works receive only, for example to watch another frequency while the first instance runs QSOs.

Transmit: the bridge touches nothing on the TX side until WSJT-X keys through it. At the first key-up it switches the radio's TX audio source to DAX, as SmartSDR's DAX button does, and puts it back when the bridge exits. It transmits at the power set in SmartSDR. It refuses to key if the radio says TX is not allowed, unkeys if WSJT-X disconnects, and never stays keyed longer than 3 minutes.

3. Without SmartSDR (standalone)

If SmartSDR is not running when the bridge starts, the bridge becomes the radio's client itself. It creates its own panadapter (band bottom + 96 kHz, e.g. 14.000–14.192) and its own TX slice on the WSJT-X dial, and tunes as WSJT-X tunes. An Antenna Genius follows that slice. Power comes from Settings, 10 W by default.

On a single-client radio such as the 8600, SmartSDR can't connect while the bridge runs standalone. Start SmartSDR first if you want both.

4. Settings (all optional)

Setting	Default	What it does
Radio	Auto	Which Flex. All radios on the LAN are listed; "Other address..." for another subnet.

Share with SmartSDR	Auto	Attach to SmartSDR when it's connected, else standalone.
Radio audio for WS	On	Radio's DAX audio (on) or the bridge's own demodulator (off).
IQ bandwidth	192 kHz	Width of IQ the radio sends to the bridge (waterfall, slices).
Linrad bandwidth	96 kHz	Width the bridge sends to QMAP. 192 kHz for WS-MAP / the improved QMAP; restart QMAP afterwards.
DAX IQ channel	1	Use 2 for a second bridge, e.g. the V polarisation.

5. If something is off

Symptom	Check
"No FlexRadio found"	The radio is on and on the same LAN. Allow Flex Bridge in Windows Firewall (the installer adds the rule), or enter the IP under Settings → Radio → Other address.
"The radio refused a GUI client"	Another program holds the radio. Start SmartSDR first and the bridge attaches to it.
QMAP frequencies wrong	QMAP: Fadd = 0, IQ sample rate Auto. The bridge sends the true centre.
QMAP waterfall empty	QMAP listens on 50004; only one program can use that port.
WSJT-X doesn't key	PTT method CAT, rig TCI Client RX1, Test CAT green.

Questions or problems: Andreas, N6NU, andreas@n6nu.org.